METRIC

DOD-STD-1399(NAVY)
SECTION 072 - PART 4
6 DECEMBER 1978

MILITARY STANDARD

INTERFACE STANDARD FOR SHIPBOARD SYSTEMS

SECTION 072 - PART 4

SHOCK ENVIRONMENT

(METRIC)

FSC 1990

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DOD-STD-1399(NAVY)
SECTION 072 - Part 4
6 December 1978

DEPARTMENT OF THE NAVY

WASHINGTON, DC 20362

Interface Standard for Shipboard Systems,
Shock Environment

DOD-STD-1399(NAVY)
SECTION 072 - Part 4

1. This Military Standard is approved for use by all interested Commands of the Department of the Navy in the technical development plans, design, and acquisition specifications for new ship acquisitions, ship modernizations or conversions, and equipment for installation therein and into active fleet ships where applicable, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Purpose. This section defines the standard interface requirements for, and the constraints on, the design of ship equipment which may be exposed to the shipboard shock environment caused by weapon explosions.

Nature of the interface. Navy ships may be subject to shock generated by non-contact underwater weapon explosions, contact weapon hits, self-delivered weapon explosions, and nuclear blast. The shock environment generated is variable both in its nature and in its effects on the performance of shipboard equipment. To reduce the potentially adverse effects of the shock environment, certain ship equipments are required to satisfy established test or design standards. The effects of shipboard shock can also be reduced through application of arrangement/installation shock-mitigating measures.

Structure. The technical content first identifies the characteristics of this environment in quantitative terms, delineating the various elements related to shock-induced displacement on board ship. Constraints with respect to shock testing requirements, and with respect to design, location, and installation factors are then established.

Numerical quantities. Numerical quantities stated in this section are expressed in metric (SI) units.

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1. GENERAL, SCOPE, INTERFACE, AND APPLICABILITY

1.1 General. The policies and procedures established by DOD-STD-1399 are mandatory. This section and the basic standard (i.e., DOD-STD-1399) are to be viewed as an integral single document.

1.2 Scope. This section establishes interface requirements for shipboard equipment susceptible to damage from shock to ensure compatibility between such equipment and the shipboard shock environment.

1.3 Interface. The basic characteristic and constraint categories concerned with this interface are shown symbolically on figure 1 (see section 3 "Definitions" of DOD-STD-1399):

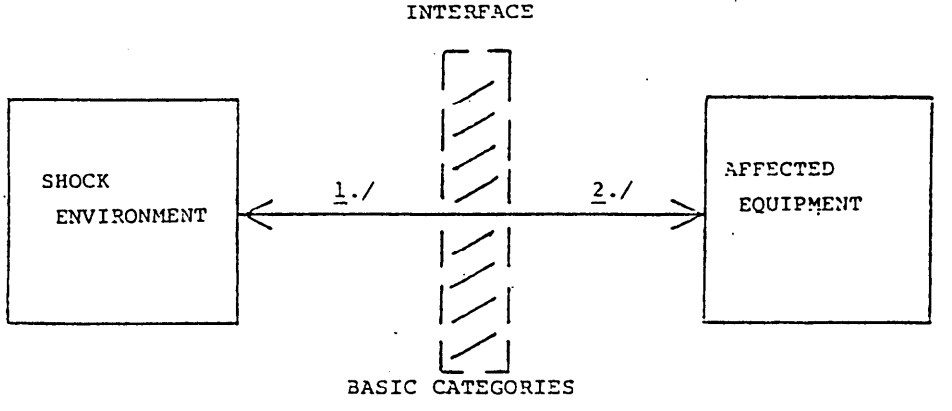


FIGURE 1. Interface.

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

1.4 Applicability. The criteria of this section are applicable to equipment intended for installation on board new ship acquisitions and modernizations or conversions. Equipment intended for installation on board active fleet ships shall possess shock resistance characteristics equal to or better than those originally required for the ship in question. The criteria of this sections do not apply to non-combatant ships, unless so specified.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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PUBLICATIONS

MILITARY

NAVAL SHIP ENGINEERING CENTER
DDS 072-1 - Shock Design Values (U).

NAVAL SEA SYSTEMS COMMAND
0908-LP-000-3010 - Shock Design Criteria for Surface Ships.

(Copies of publications may be obtained from the Naval Ship Engineering Center, SEC 6105, Department of the Navy, Washington, DC 20362. DDS 072-1 requires that contracting activities shall certify the need-to-know and security clearance data of the contractor or bidder and forward such certification to the above address authorizing release of that document.)

3. DEFINITIONS

3.1 Shock environment. The shock environment is the physical result of non-contact explosions, contact explosions, or nuclear blasts which impart rapid displacement, velocity, and acceleration to the ship structure.

3.2 Equipment grade designations.

3.2.1 Grade A. Equipment essential for the safety and continued combat capability of the ship. Such equipment shall withstand shock loadings with no significant effect upon performance and without any portion of the equipment coming adrift or otherwise creating a hazard to personnel or vital systems.

3.2.2 Grade B. Equipment not required for the safety or continued combat capability of the ship. Such equipment shall withstand shock loadings without the equipment or any external portion coming adrift or otherwise creating a hazard to personnel or vital systems.

3.2.3 Grade C. No shock test or shock design requirements.

4. REQUIREMENTS

4.1 Requirements. The specific interface requirements and constraints established hereby are mandatory and shall be adhered to by SYSCOMS, Project managers, contractors, and all others engaged in any aspect of shipboard design to which these requirements and constraints apply, including equipment design, production, and installation (see section 4 "Requirements" of DOD-STD-1399).

5. INTERFACE CHARACTERISTICS AND CONSTRAINTS

5.1 General considerations. The effect of shock on a ship and its installed equipment depends upon many complex factors. The major factors involved are the proximity of the explosion, the mass of the charge or weapon yield, and the medium in which the explosion occurs. There are many factors affecting the intensity of the shock to which a particular item of shipboard equipment is subjected, among them the construction of the hull, the location of the equipment, its foundation, and its mass. For a given ship, the sum of these factors determine whether no damage is sustained, shock damage to equipment occurs, or substantial hull and equipment damage results. Many shock patterns and mode couplings are produced, and these differ at various locations due to the ship's structural configuration. When subjected to the shock environment, the ship structure can be physically displaced in any direction. This displacement is transferred to installed equipment through foundations, mounting brackets, or other securing devices. As a result of structural displacement, a velocity is imparted to installed equipment and, as a result of this velocity, acceleration is imparted to the equipment.

5.1.1 Range of shock magnitude. The ranges of magnitude of the effects of the shock environment depend on the displacement/time relationship $X = F(t)$; i.e., displacement is a function of time. The time frame of interest for an empirical "worst case" lies in the range from zero to a few milliseconds; velocity values range from zero to high orders of magnitude. Figure 2 demonstrates the basic physical and mathematical relationship between displacement, velocity, and acceleration.

5.1.2 Minimizing shock effects. The primary goal is to provide maximum assurance that the ship can accomplish its mission when operated under combat conditions. Environmental shock effects can be minimized by design of equipment and foundations, and by judicious location of the equipment within the ship. Equipment, in accordance with its grade designation, will be designed to operate in the postulated shock environment.

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5.1.3 Simulation. It is impossible to precisely define or postulate the shock environment to which a ship or submarine may be subjected under combat conditions. Therefore, reliance is placed upon an empirical testing program which simulates realistic conditions to the maximum degree practical (see 5.2.1). This is usually accomplished by the use of shock testing which subjects equipment to conditions which simulate actual shock effects. Compatibility of tested equipment with the shock environment is established if the equipment meets the requirements of the grade of shock hardening specified (see 3.2). Shock design requirements in lieu of testing are applied to items which are too large or heavy to permit shock testing. Shipboard installation environmental criteria also apply (see 5.3.5).

5.2 Interface characteristics. The interface characteristics of the ship's shock environment are specified in 5.2.1. These characteristics, to varying degrees, will be present on all ships when subjected to a shock environment, and will have an effect on equipment installed therein. Such characteristics are described qualitatively since quantitative values for particular shipboard situations are not readily obtainable, and the ultimate installed location of the equipment may not be known. Therefore, general empirical criteria of the shock environment related to "worst case" conditions which may reasonably be expected have been established.

5.2.1 Postulated "worst case". Empirical "worst case" shock environment conditions for design purposes must be postulated. This "worst case" condition cannot be demonstrated directly, but must be simulated. The test requirements of MIL-S-901 provide such simulation. Design shock loadings are specified in DDS 072-1, and dynamic design procedures are contained in NAVSEA 0908-LP-000-3010 (see 5.3.2).

5.3 Interface constraints. The interface characteristics of the shipboard shock environment impose certain constraints on the design, testing, location, and installation of equipment exposed to the shock environment. These constraints are specified in 5.3.1 through 5.3.5.

5.3.1 Compatibility. The design of equipment exposed to the shipboard shock environment shall be compatible with the interface characteristics given in 5.2.

5.3.2 Equipment design. Equipment shall be designed to withstand the postulated shock environment in accordance with the degree of shock hardening specified (grade A or grade B (see 3.2)) and testing as specified by MIL-S-901.

5.3.3 Location criteria. To the maximum extent practicable, units of equipment shall be located and arranged within the ship so as to minimize shock effects.

5.3.4 Installation considerations. Units of equipment shall be installed so that:

(a) Sufficient clearance is provided to prevent the unit from striking adjacent structures or other objects during maximum excursions.
(b) If resiliently mounted, flexible connections are used. Other installation considerations shall be in accordance with applicable specifications.

5.3.5 Test procedures and methods. Test procedures for qualification testing of units to withstand high-impact shock are specified in MIL-S-901. These procedures are generalized to cover all ship systems and equipments and thus do not address the specific installation details for any particular equipments. Equipment which is being shock tested shall be installed for tests in the same manner as it will be installed aboard ship; e.g., resiliently or solidly mounted, base mounted only, and base and back mounted, etc.; however, when mounting options exist, it is preferred that units be mounted using one mounting plane, i.e., base or back mounted, but not base and back. This precludes limitation of locations where equipment can be suitably installed on board, and obviates the problem of shock-induced relative motion between the points of attachment.

5.3.6 Technical information and guidance. The Naval Ship Engineering Center (NAVSEC), Ship Survivability Office, will provide additional technical information and engineering guidance upon request.

6. DOCUMENTATION REQUIREMENTS

6.1 DD Form 1423. When this standard is invoked in direct Government acquisition, or on a prime contractor who must obtain the data for contractor-furnished equipment (CFE) from his subcontractor, the following data requirements, as applicable, shall be specified by the Principal Development Activity (PDA) on DD Form 1423 (Contract Data Requirements List)

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attached to the contract or order. Data Item Description UDI-T-23753 may be cited in the DD Form 1423, blocks 2 and 4, as appropriate, with suitable further identification of the particular data item in block 3. The NAVSEC, Ship Survivability Office, shall be included in the distribution listed in block 14 of DD Form 1423 for data specified.

(a) Test reports as prescribed in MIL-S-901 for high impact shock.
(b) Other, as specified.

7. DEVIATIONS

7.1 Deviation policy. In achieving the purpose of this section, it is recognized that there must be some flexibility of application (see section 6 "Deviations" of DOD-STD-1399). The specific policy for preparation, review, and approval of requests to deviate from the shock requirements specified herein are as follows:

(a) Deviation requests shall be prepared by the Navy office or activity responsible for acquiring the item in question, and shall include the information indicated in 7.1.1.
(b) A copy of all Deviation requests shall be furnished to the NAVSEC, Ship Survivability Office (SEC 6105) for technical review.
(c) Naval Sea Systems Command - designated Project Managers (PMS's) and Ship Logistics Managers (SLM's) shall take approval action on Deviation requests under their cognizance, with consideration or recommendations furnished by SEC 6105. For purposes of this requirement, deviation requests are considered to be under PMS or SLM cognizance only in cases where the PMS or SLM possesses the authority to modify the contract for the development or acquisition of the item for which the deviation is sought.
(d) Deviation requests not under PMS or SLM cognizance as defined above shall be forwarded to the Commander, Naval Sea Systems Command for approval, accompanied by SEC 6105 recommendations.
(e) In cases where the deviation request is approved, the following shall be provided copies of the request (with accompanying SEC 6105 recommendations) and approval correspondence:

(1) COMNAVSEA (if approval was granted by a PMS or SLM).
(2) SEC 6105.
(3) SEA 06N.
(4) CHNAVMAT (MAT 08).

7.1.1 Deviation request content. The following data shall be included in all deviation requests:

(a) Description of item. Identify the item, provide a basic physical description of the item (including size, mass, and mounting arrangements), and briefly describe functions performed by the item.
(b) Potential consequences of shock damage. Assess the potential effects of loss of item function upon essential mission capabilities. In cases where the item is known to suffer from specific shock deficiencies (as revealed by shock design studies or previous shock testing), describe the deficiencies and assess effects. Describe potential effects upon mission capabilities immediately after attack (prior to reaction by ship's force), and probable long-term effects (considering availability/capabilities of back-up systems and potential for repair by ship's force). Indicate and rationalize whether these effects are considered to be of critical, high, medium, or low importance to the survivability and mission keeping capability of the ship. Identify affected ships.
(c) Justification for deviation from shock requirements. State reasons why deviation from the shock criteria contained herein is recommended. The fact that the item has performed satisfactorily during normal shipboard service shall not be cited as justification for deviation from shock requirements. In cases where impracticality of modifying existing designs is cited as a basis for deviation, the feasibility of using shock-isolating mountings to protect existing equipment against shock shall be evaluated.
(d) Planned corrective action. State plans and schedules for future shock test or shock design evaluation of the item, or for future shock hardening and re-evaluation of items which have previously proven deficient from a shock standpoint. If no future corrective action is planned or being budgeted for, state reasons.

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(e) Cognizant office. Identify the Navy office responsible for furnishing the item.

Note: Where appropriate, a single deviation request may be submitted to cover a category or family of functionally similar items.

7.1.2 Deviation procedure. Requests for deviations shall be submitted to the Naval Sea Systems Command with copies to:

(a) Program/Project manager.
(b) NAVSEC 6105.

Review activity:
EC
User activity:
AS

Preparing activity:
Navy - SH
(Project 1990-N014)

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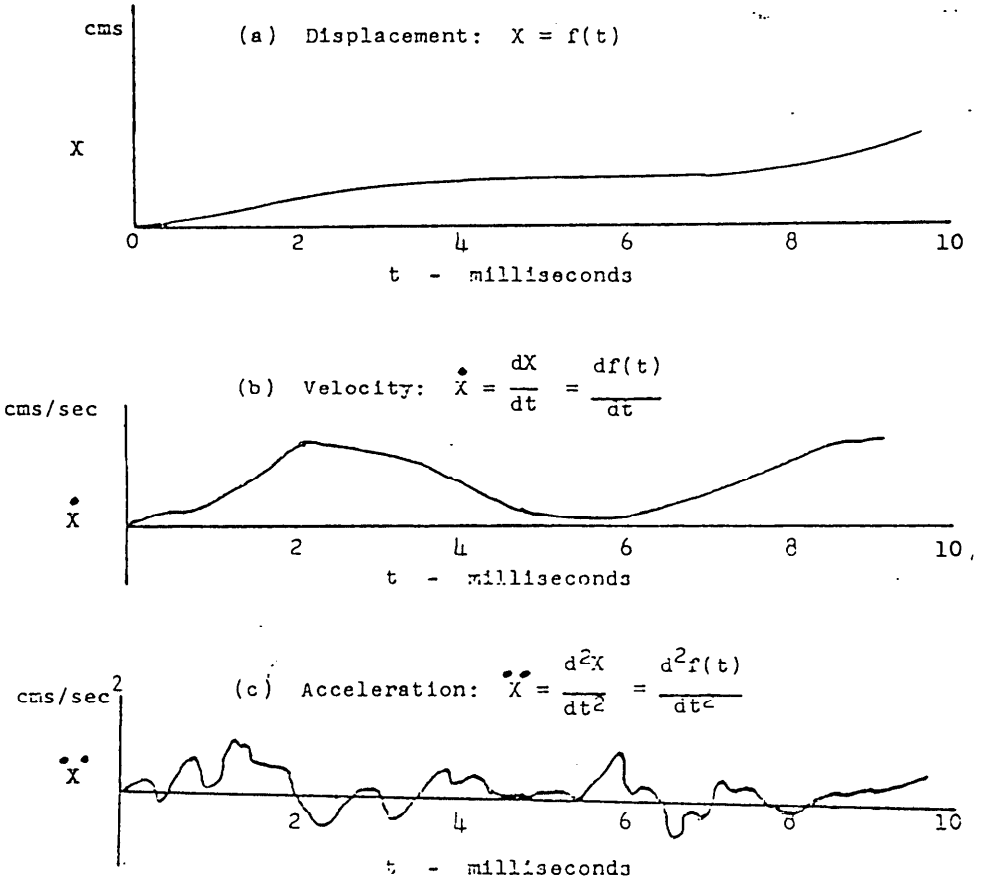


FIGURE 2. Example of a typical shock curve.

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